

A Comparison of SD Card Energy Consumption

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This application note describes an update to the AudioMoth firmware (AudioMoth-Firmware-Basic 1.12.0 and AudioMoth-GPS-Sync 1.2.3) that reduces energy consumption with larger capacity SD cards.

1 SD Card Energy Consumption

SD cards that are used with AudioMoth consume no energy between recordings as they are switched off by the AudioMoth hardware. However, during recordings, they consume energy both when actually writing data to storage, and also while sleeping between writes.

Most small SD cards (those with a capacity of less than 32 GB) will automatically sleep between writes, consuming very little energy. Their signal to go to sleep is the state of the chip select (CS) line which goes low only when a write operation is required, and is high otherwise. However, larger SD cards often do not behave in this way.

Figure 1 shows a comparison of the energy consumption (in mA) of an AudioMoth using three different families of SD cards, Kingston Canvas Select Plus, Samsung EVO Plus and SanDisk Extreme, with sizes of 32, 64, 128, 256 and 512 GB, making recordings at a sample rate of 48 kHz. The SD card writes consists of packets of 32768 bytes, and at 48 kHz occur every 341 ms. The baseline energy consumption of approximately 9 mA is determined by AudioMoth itself. Upon which the SD card sleep and write current is added. Note that many of the larger SD cards do not sleep between writes, but continue to consume significant amounts of energy (shown in red).

To address this behaviour, updated versions of the AudioMoth firmware (specifically AudioMoth-Firmware-Basic 1.12.0 and AudioMoth-GPS-Sync 1.3.0 onwards) also pauses the SD card clock between writes. This provides an additional signal to the SD card that it should go to sleep. Figure 2 shows the same SD card energy measurements when the SD card clock is paused. Note that now all of the SD cards sleep as required. The slight exception are the larger SanDisk Extreme cards (128, 256 and 512 GB) which do eventually sleep but

only after a further delay of 100 ms.

Figures 3 and 4 show the same comparison for three families of endurance SD cards: Kingston High Endurance, Samsung Pro Endurance and SanDisk High Endurance, with sizes of 128 and 256 GB, again making recordings at a sample rate of 48 kHz. Pausing the SD card clock between writes makes a significant difference in energy consumption as all the cards now sleep between writes.

2 Sample Rate

The total energy consumption increases with sample rate as the AudioMoth micro-controller itself spends less time sleeping between audio capture events, and the SD card writes occur more frequently. Table 1 shows the average energy consumption (in mA) whilst making recordings at all possible sample rates. For sample rates below 96 kHz, the standard energy consumption is reported, along with that when Energy Saver Mode (ESM) used. In this mode, the frequency of the AudioMoth micro-controller clock is divided by two, and less over-sampling is performed, in order to reduce overall energy consumption. Also shown are the reference values used in the AudioMoth Configuration App (from version 1.13.0 onwards)¹. These represent rounded average values for the lower energy cards.

3 Discussion

The energy consumption of SD card varies significant across makes, models and sizes. Where the maximum deployment lifetime is most important, we recommend measuring the energy consumption of the AudioMoth with the card to be used. However, in these experiments, Kingston Canvas Select Plus and Samsung Pro Endurance cards have demonstrated the lowest energy consumption in their class.

¹Prior versions of the AudioMoth Configuration App used the values for a SanDisk Extreme 32 GB card.

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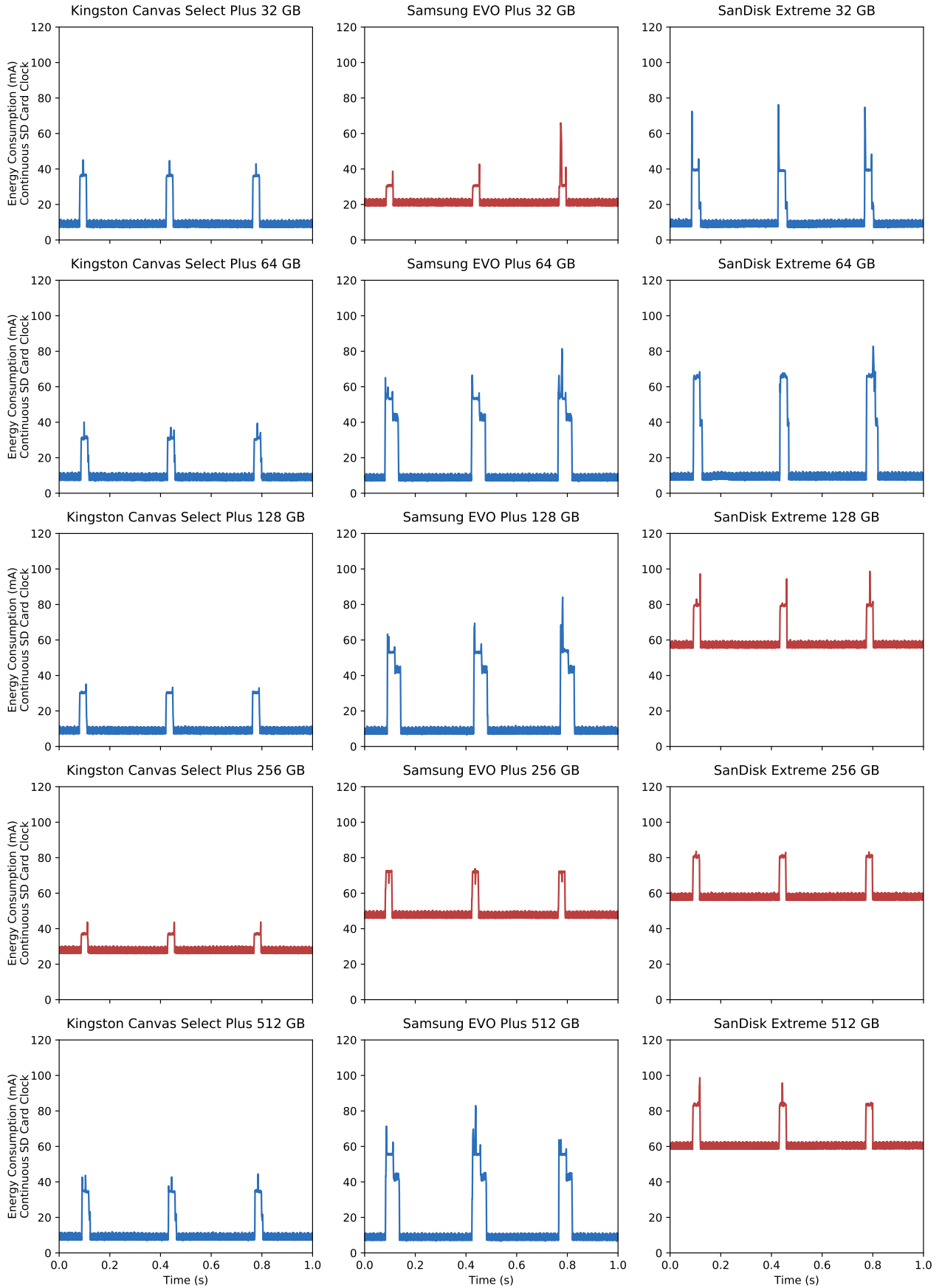


Figure 1: Energy consumption of AudioMoth using multiple SD cards when recording at 48 kHz using the previous versions of the AudioMoth firmware which do not pause the SD clock between writes.

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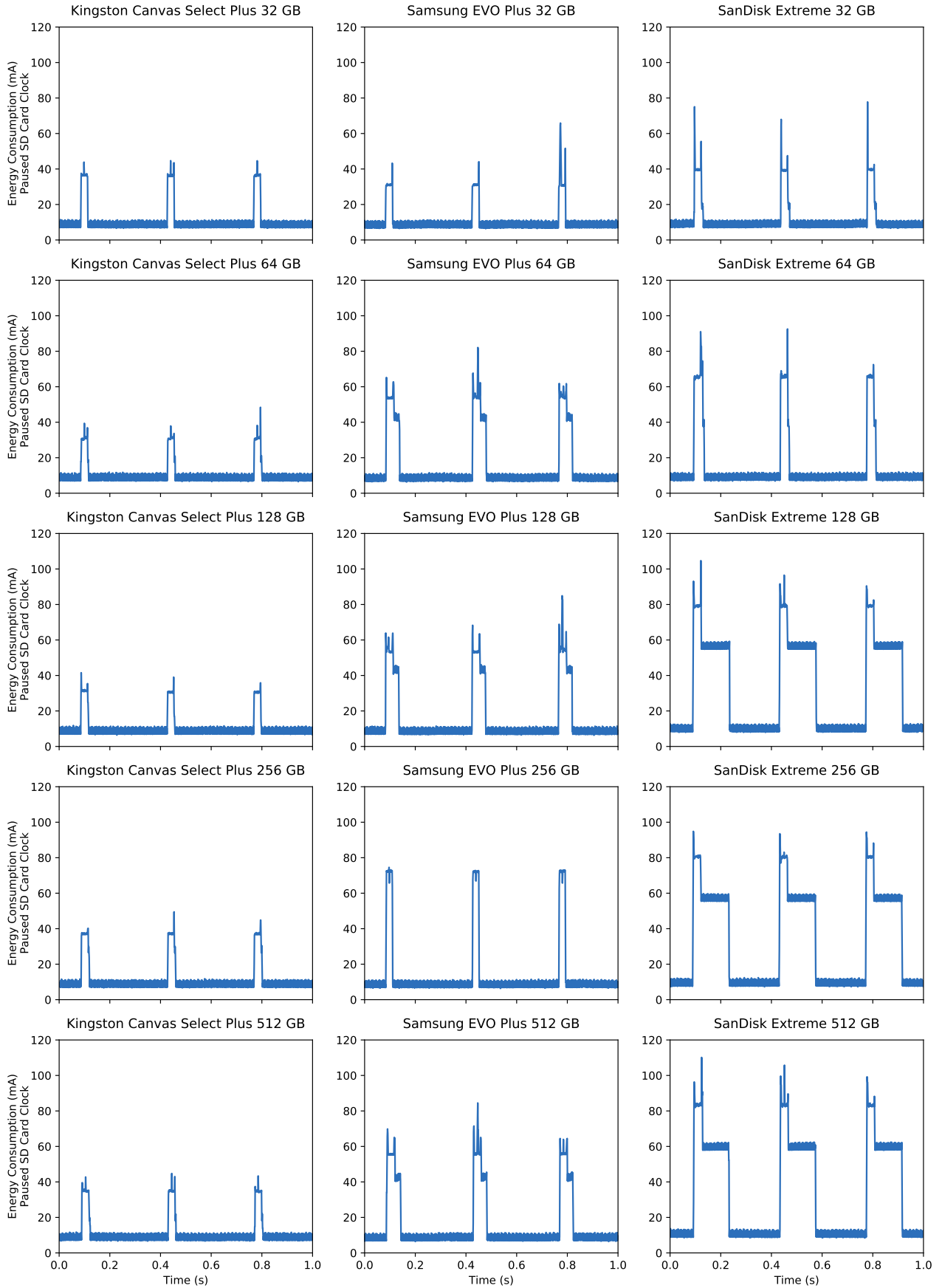


Figure 2: Energy consumption of AudioMoth using multiple SD cards when recording at 48 kHz using the updated version of the AudioMoth firmware which pauses the SD clock between writes.

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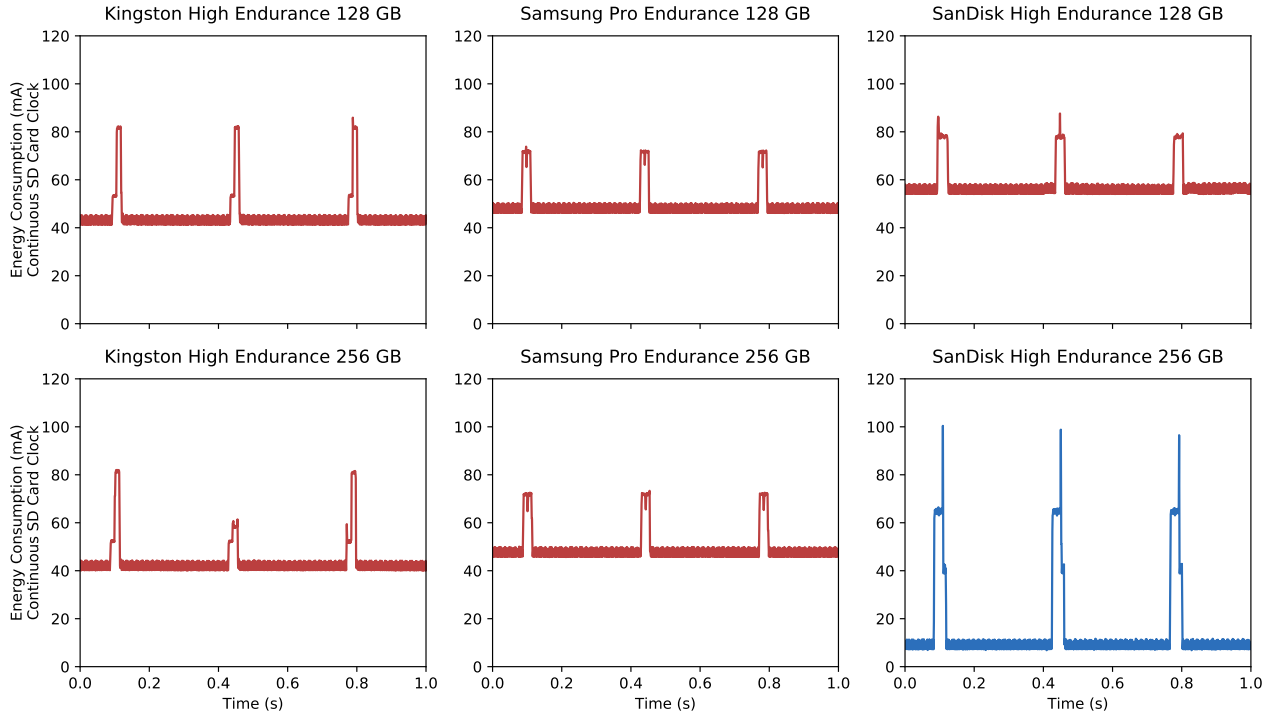


Figure 3: Energy consumption of AudioMoth using multiple SD cards when recording at 48 kHz using the previous versions of the AudioMoth firmware which do not pause the SD clock between writes.

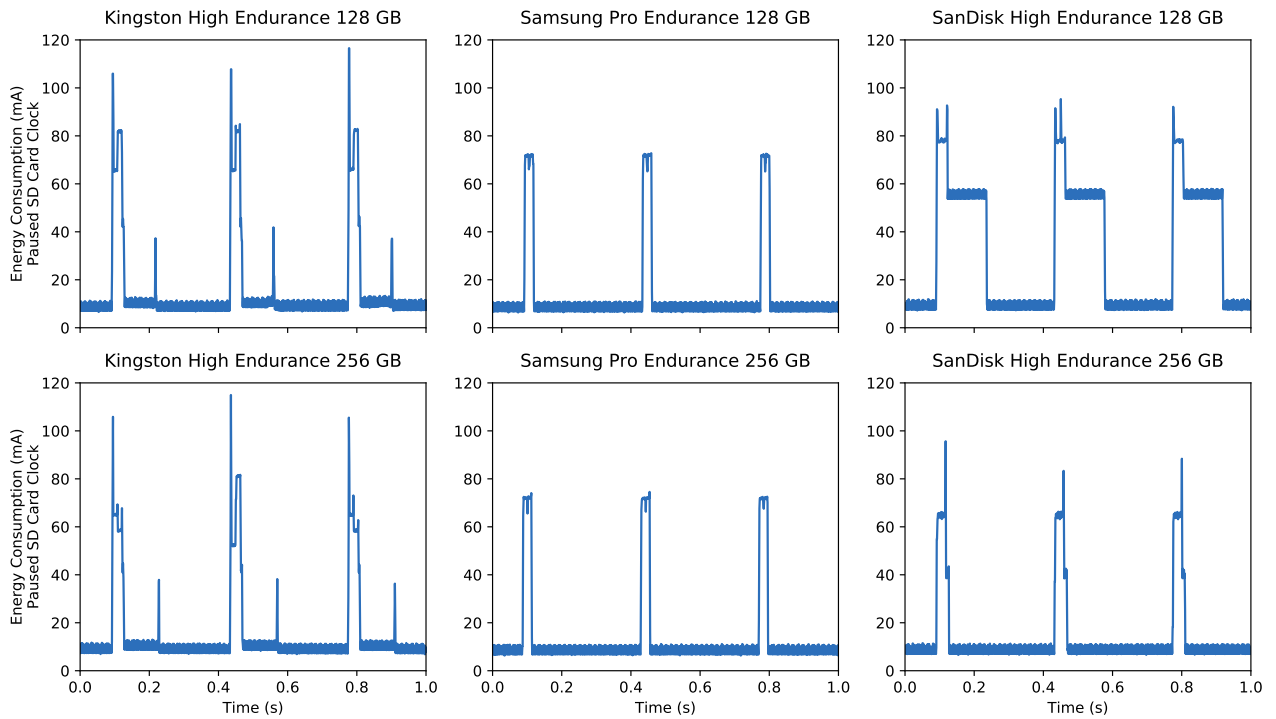


Figure 4: Energy consumption of AudioMoth using multiple SD cards when recording at 48 kHz using the updated version of the AudioMoth firmware which pauses the SD clock between writes.

	8 kHz		16 kHz		32 kHz		48 kHz		96 kHz	192 kHz	250 kHz	384 kHz
	ESM	—	ESM	—	ESM	—	ESM	—				
Kingston Canvas Select Plus 32 GB	5.5	8.4	5.8	8.8	6.7	9.7	7.6	10.7	13.8	21.9	16.8	25.5
Kingston Canvas Select Plus 64 GB	5.4	8.4	5.8	8.8	6.5	9.5	7.2	10.3	12.8	18.3	16.1	22.1
Kingston Canvas Select Plus 128 GB	5.3	8.3	5.7	8.7	6.4	9.4	7.1	10.2	12.6	17.8	15.6	21.3
Kingston Canvas Select Plus 256 GB	5.5	8.5	6.0	9.1	7.1	10.0	8.1	11.1	14.5	21.6	19.1	26.9
Kingston Canvas Select Plus 512 GB	5.5	8.5	6.0	9.0	6.8	9.8	7.7	10.8	13.8	20.0	17.5	24.3
Kingston Endurance 128 GB	6.1	9.4	7.4	10.5	10.0	12.7	12.5	17.2	25.3	32.5	28.9	42.6
Kingston Endurance 256 GB	6.4	9.3	7.9	10.7	10.2	13.4	12.9	15.9	22.6	34.5	27.7	40.4
Samsung EVO Plus 32 GB	5.3	8.2	5.6	8.6	6.4	9.4	7.3	10.4	13.1	18.8	16.9	23.3
Samsung EVO Plus 64 GB	6.0	9.0	7.2	10.2	9.5	12.4	11.7	14.7	21.8	36.4	34.9	52.1
Samsung EVO Plus 128 GB	6.0	9.0	7.2	10.2	9.5	12.4	11.7	14.7	21.8	37.0	35.4	52.7
Samsung EVO Plus 256 GB	5.8	8.8	6.7	9.8	8.7	11.6	10.7	13.5	19.6	33.4	25.7	39.9
Samsung EVO Plus 512 GB	6.1	9.1	7.3	10.3	9.7	12.6	11.9	14.9	22.1	37.4	35.6	53.2
Samsung Pro Endurance 128 GB	5.8	8.8	6.7	9.7	8.5	11.4	10.3	13.2	19.0	31.9	24.7	37.7
Samsung Pro Endurance 256 GB	5.7	8.7	6.6	9.6	8.3	11.3	10.3	13.2	19.0	31.9	25.0	38.0
SanDisk Extreme 32 GB	5.8	8.8	6.6	9.6	7.5	10.5	8.5	11.7	15.2	23.5	21.0	30.1
SanDisk Extreme 64 GB	6.2	9.1	7.1	10.1	9.0	12.0	10.9	14.0	20.0	32.8	27.3	41.6
SanDisk Extreme 128 GB	10.0	12.9	13.6	16.4	21.2	24.2	28.6	31.7	54.8	67.1	64.8	70.7
SanDisk Extreme 256 GB	9.4	12.4	13.2	16.2	20.7	23.7	28.2	31.4	54.6	67.7	65.5	71.4
SanDisk Extreme 512 GB	10.6	13.6	14.4	17.4	22.0	25.0	29.6	32.8	56.4	70.8	68.6	75.1
SanDisk High Endurance 128 GB	9.2	12.1	12.9	15.7	20.3	22.9	27.7	30.8	53.6	65.6	63.4	69.3
SanDisk High Endurance 256 GB	6.2	9.1	7.2	10.2	9.1	12.2	11.1	14.2	20.4	34.8	29.7	43.1
AUDIOMOTH CONFIGURATION APP	5.5	8.5	6.0	9.0	7.0	10.0	7.5	10.5	13.5	20.0	17.0	24.0

Table 1: Average consumption of AudioMoth using multiple SD cards when recording at different sample rates using the updated version of the AudioMoth firmware which pauses the SD clock between writes. Note that ESM represents the use of Energy Saver Mode, selected in the AudioMoth Configuration App.